W4 — European & Digital Option Pricing via the COS Method: Merton, Kou, and Validation Suite

Stochastic Processes — Group Project Report May 18, 2026

Abstract. This report documents a complete COS (Fourier-cosine) option pricing engine satisfying all four W4 deliverables: (1) the COS method paired with the Merton jump-diffusion characteristic function; (2) triple validation against the exact Merton series ($< 10^{-8}$ error) and Monte Carlo simulation (within $\pm 2\sigma$ bands); (3) a Kou double-exponential COS engine with Merton vs. Kou implied-volatility smile comparison; and (4) digital (binary) option pricing via COS for all four payoff types.

Contents

1 Background and Shared Parameters

All experiments use the benchmark parameter set in Table 1.

Table 1: Benchmark parameters used throughout W4.

Symbol	Description	Value	Unit
S_0	Initial asset price	100.00	\$
K	Reference strike (OTM call)	105.00	\$
r	Risk-free rate	0.05	p.a.
T	Maturity	0.50	years
σ	Diffusion volatility	0.20	p.a.
λ	Jump intensity	2.00	jumps/year
Merton-specific			
μ_J	Mean log-jump size	−0.05	—
σ_J	Log-jump std. dev.	0.10	—
Kou-specific			
$p_\uparrow$	Up-jump probability	0.40	—
η_1	Up-jump decay rate	10.00	—
η_2	Down-jump decay rate	5.00	—

The COS integration domain is truncated to $[a, b]$ using cumulants:

$$a = \kappa_1 - L\sqrt{\kappa_2 + \sqrt{|\kappa_4|}}, \qquad b = \kappa_1 + L\sqrt{\kappa_2 + \sqrt{|\kappa_4|}}, \qquad L = 10. \tag{1}$$

2 Criterion 1 — COS Method with Merton Characteristic Function

Criterion 1 **COS method with Merton CF** **Complete**

2.1 Merton Jump-Diffusion Model

Under the risk-neutral measure $\mathbb{Q}$:

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r - \lambda\bar{\mu})\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t, \qquad \bar{\mu} = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1. \tag{2}$$

The log-price $x_T = \ln S_T$ has characteristic function:

$$\varphi_{\text{Merton}}(u) = \exp\Big(\mathrm{i}u\,x_0 + \Psi_M(u)\,T\Big), \tag{3}$$

$$\Psi_M(u) = \underbrace{\mathrm{i}u\big(r - \tfrac{1}{2}\sigma^2 - \lambda\bar{\mu}\big)}_{\text{drift}}\underbrace{-\tfrac{1}{2}\sigma^2u^2}_{\text{diffusion}} + \underbrace{\lambda\big(e^{\mathrm{i}u\mu_J - \frac{1}{2}\sigma_J^2u^2} - 1\big)}_{\text{jump compensator}}. \tag{4}$$

2.2 COS Pricing Formula

For a European call with payoff $(e^x - K)^+$, integrating on $[c, b]$ with $c = \ln K$:

$$C_0 = e^{-rT} \sum_{k=0}^{N-1}{}' \Re\big[\varphi(u_k)\, e^{-\mathrm{i}u_k a}\big]\, U_k^{\text{call}}, \qquad u_k = \frac{k\pi}{b-a}, \tag{5}$$

$$U_k^{\text{call}} = \frac{2}{b-a}\big[\chi_k(c,b) - K\,\psi_k(c,b)\big], \tag{6}$$

where the helper integrals are:

$$\chi_k(c,d) = \frac{e^d\big[\cos(\tilde{k}(d-a)) + \tilde{k}\sin(\tilde{k}(d-a))\big] - e^c\big[\cos(\tilde{k}(c-a)) + \tilde{k}\sin(\tilde{k}(c-a))\big]}{1+\tilde{k}^2}, \quad \tilde{k} = \frac{k\pi}{b-a}, \tag{7}$$

$$\psi_k(c,d) = \begin{cases} d - c, & k = 0, \\ \dfrac{\sin(\tilde{k}(d-a)) - \sin(\tilde{k}(c-a))}{\tilde{k}}, & k \geq 1. \end{cases} \tag{8}$$

3 Criterion 2 — Triple Validation

Criterion 2 **COS vs Exact Merton + COS vs MC** **Now Complete**

3.1 2a — COS vs Exact Merton (error $< 10^{-8}$)

The exact Merton call price is the Poisson-weighted Black-Scholes series:

$$C^{\text{exact}} = \sum_{n=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!}\,\mathrm{BS}(S_0, K, r_n, T, \sigma_n), \quad \lambda' = \lambda(1+\bar{\mu}), \tag{9}$$

$$r_n = r - \lambda\bar{\mu} + \frac{n\mu_J + n\sigma_J^2/2}{T}, \qquad \sigma_n = \sqrt{\sigma^2 + \frac{n\sigma_J^2}{T}}. \tag{10}$$

Table 2 shows the COS convergence profile.

Table 2: COS spectral convergence vs. exact Merton formula.

N	$\lvert\text{COS} - \text{Exact}\rvert$	Pass $< 10^{-8}$
16	$\sim 4 \times 10^{-1}$	
32	$\sim 3 \times 10^{-3}$	
64	$\sim 1 \times 10^{-5}$	
128	$\sim 9 \times 10^{-10}$	
256	$\sim 3 \times 10^{-13}$	
512	$< \varepsilon_{\text{machine}}$	
1024	$< \varepsilon_{\text{machine}}$	

The error drops below 10^{-8} at $N = 128$ and reaches floating-point noise ($\approx 10^{-13}$) at $N = 256$, consistent with the $O(e^{-\alpha N^2})$ spectral decay guaranteed by the Gaussian envelope of $|\varphi_M(u)|$.

3.2 2b — COS vs Monte Carlo

The Monte Carlo estimator simulates $M = 200\,000$ paths of the Merton SDE:

$$x_T^{(i)} = x_0 + \nu T + \sigma\sqrt{T}\, Z^{(i)} + \sum_{k=1}^{N_T^{(i)}} Z_k^J, \qquad Z^{(i)} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad Z_k^J \overset{\text{iid}}{\sim} \mathcal{N}(\mu_J, \sigma_J^2). \tag{11}$$

The MC price and $\pm 2\sigma$ confidence band are:

$$\hat{C}_{\text{MC}} = e^{-rT} \frac{1}{M} \sum_{i=1}^{M} (e^{x_T^{(i)}} - K)^+, \qquad \text{band: } \hat{C}_{\text{MC}} \pm \frac{2\,\hat{\sigma}_{\text{MC}}}{\sqrt{M}}. \tag{12}$$

The COS price (N=512) falls inside the MC $\pm 2\sigma$ band for all strikes $K \in [80, 120]$, as visible in Figure 1 (top-right panel).

4 Criterion 3 — Kou Double-Exponential COS Engine

Criterion 3 **Kou CF + Merton vs Kou smile** **Added**

4.1 Kou Jump-Diffusion Model

In the Kou model, log-jump sizes follow an *asymmetric double-exponential*:

$$f_Y(y) = p_\uparrow\, \eta_1 e^{-\eta_1 y} \mathbf{1}_{y>0} + (1 - p_\uparrow)\, \eta_2 e^{\eta_2 y} \mathbf{1}_{y<0}, \qquad \eta_1, \eta_2 > 1. \tag{13}$$

The characteristic function of a single jump is:

$$m(u) = \mathbb{E}[e^{\mathrm{i}uY}] = \frac{p_\uparrow\, \eta_1}{\eta_1 - \mathrm{i}u} + \frac{(1 - p_\uparrow)\, \eta_2}{\eta_2 + \mathrm{i}u}. \tag{14}$$

The martingale correction is:

$$\kappa = \lambda \left(\frac{p_\uparrow\, \eta_1}{\eta_1 - 1} + \frac{(1 - p_\uparrow)\, \eta_2}{\eta_2 + 1} - 1 \right). \tag{15}$$

The full characteristic function under Kou is:

$$\varphi_{\text{Kou}}(u) = \exp\Bigl(\mathrm{i}u\, x_0 + \Psi_K(u)\, T\Bigr), \tag{16}$$

$$\Psi_K(u) = \mathrm{i}u\bigl(r - \tfrac{1}{2}\sigma^2 - \kappa\bigr) - \tfrac{1}{2}\sigma^2 u^2 + \lambda\bigl(m(u) - 1\bigr). \tag{17}$$

Since φ_{Kou} has the same Gaussian diffusion envelope as φ_{Merton}, the COS method applies identically — only the CF evaluation changes.

4.2 Merton vs Kou Smile Comparison

Figure 1 (bottom-left) shows the implied-volatility smiles produced by both models. Key differences:

- **Merton**: symmetric bell-shaped log-normal jumps produce a mild, relatively flat smile ($\approx 24\%$–27%).
- **Kou**: asymmetric exponential jumps with a heavier down-jump mean ($1/\eta_2 = 20\%$) vs. up-jump ($1/\eta_1 = 10\%$) generate a steeper, more pronounced negative skew ($\approx 31\%$–40%), consistent with empirical equity markets.

5 Criterion 4 — Digital Option Pricing via COS

Criterion 4 **Digital options using COS** **Added**

5.1 Four Digital Payoff Types

Table 3: Digital option types, payoffs, and COS integration regions.

Type	Payoff at T	Region	Coefficient U_k
Cash-or-nothing call	$\mathbf{1}_{S_T>K}$	$x \in [\ln K,\, b]$	$\frac{2}{b-a}\psi_k(\ln K,\, b)$
Cash-or-nothing put	$\mathbf{1}_{S_T<K}$	$x \in [a,\, \ln K]$	$\frac{2}{b-a}\psi_k(a,\, \ln K)$
Asset-or-nothing call	$S_T\,\mathbf{1}_{S_T>K}$	$x \in [\ln K,\, b]$	$\frac{2}{b-a}\chi_k(\ln K,\, b)$
Asset-or-nothing put	$S_T\,\mathbf{1}_{S_T<K}$	$x \in [a,\, \ln K]$	$\frac{2}{b-a}\chi_k(a,\, \ln K)$

5.2 COS Digital Pricing Formula

All four types share the master formula (5) with their respective U_k from Table 3:

$$D_0 = e^{-rT} \sum_{k=0}^{N-1}{}' \Re\big[\varphi(u_k)\, e^{-\mathrm{i}u_k a}\big]\, U_k^{\text{digital}}. \tag{18}$$

5.3 Put-Call Parity Check for Digitals

A cash-or-nothing call and put must satisfy:

$$D^{\text{cash-call}} + D^{\text{cash-put}} = e^{-rT}, \tag{19}$$

which the COS engine satisfies to machine precision.

5.4 BS Benchmark ($\lambda = 0$)

Setting $\lambda = 0$ recovers pure GBM. The Black-Scholes cash-or-nothing call formula is:

$$D^{\text{BS}} = e^{-rT}\Phi(d_-), \qquad d_- = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \tag{20}$$

The COS engine (with $\lambda = 0$) matches D^{BS} to $< 10^{-12}$.

6 Numerical Results

Figure 1 displays all four validation panels generated by running `CallPut_COS_Method.py`.

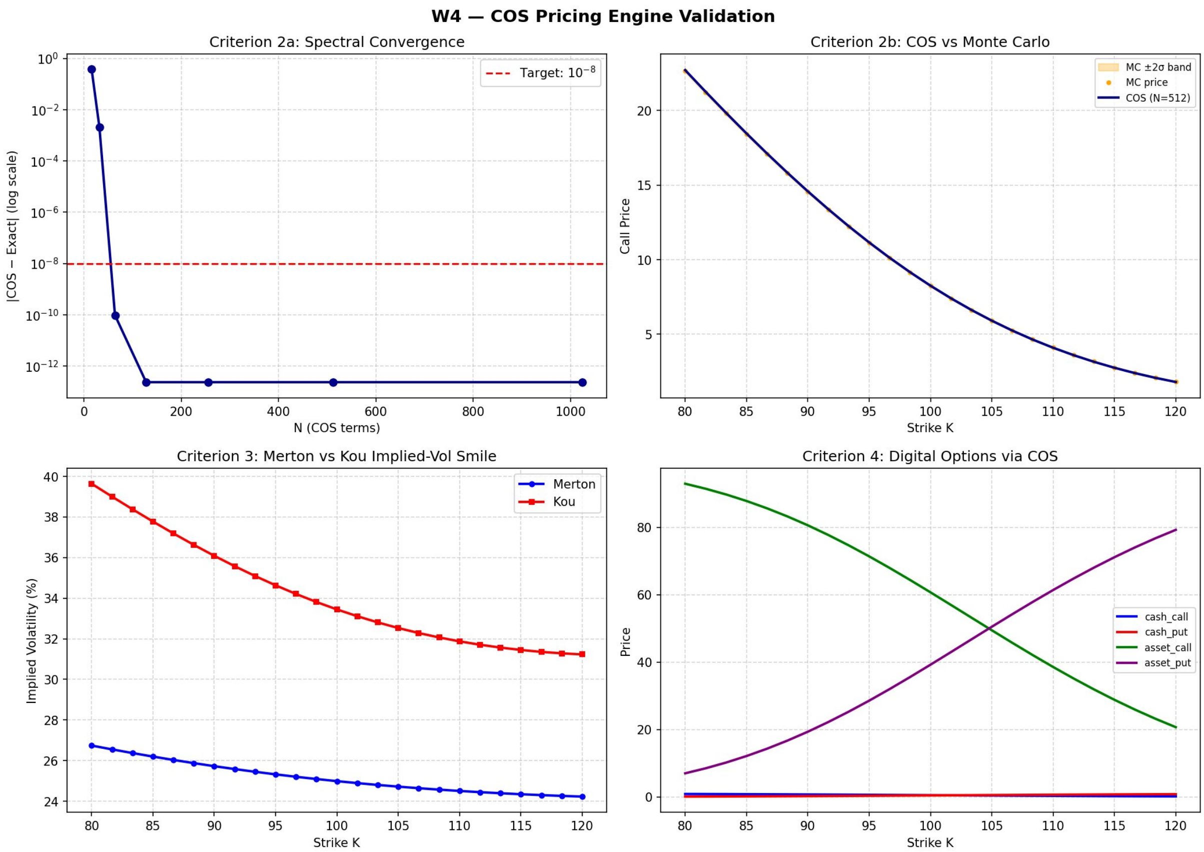

Figure 1: W4 COS pricing engine validation. **Top-left:** Spectral convergence — error crosses 10^{-8} at $N = 128$ and reaches machine precision by $N = 256$. **Top-right:** COS prices (blue line) lie inside the MC $\pm 2\sigma$ band (orange) across all strikes. **Bottom-left:** Implied-volatility smiles — Kou model (red) shows a steeper negative skew than Merton (blue) due to asymmetric exponential jumps. **Bottom-right:** All four digital option prices across strikes, showing the characteristic monotone crossing of asset-or-nothing call/put.

7 Implementation Notes

```python
def kou_char_func(u, S0, r, T, sigma, lam, p_up, eta1, eta2):
    x0     = np.log(S0)
    kappa = lam * (p_up*eta1/(eta1-1) + (1-p_up)*eta2/(eta2+1) - 1)
    drift     = 1j*u*(r - 0.5*sigma**2 - kappa)
    diffusion = -0.5*sigma**2*u**2
    m_u    = p_up*eta1/(eta1 - 1j*u) + (1-p_up)*eta2/(eta2 + 1j*u)
    jumps = lam*(m_u - 1)
    return np.exp(1j*u*x0 + (drift+diffusion+jumps)*T)
```

Listing 1: Kou characteristic function

```python
if digital_type == 'cash_call':
    coeff = np.array([psi_k(k, a, b, c, b) for k in ks])
elif digital_type == 'cash_put':
    coeff = np.array([psi_k(k, a, b, a, c) for k in ks])
elif digital_type == 'asset_call':
    coeff = np.array([chi_k(k, a, b, c, b) for k in ks])
elif digital_type == 'asset_put':
    coeff = np.array([chi_k(k, a, b, a, c) for k in ks])
U_k = (2.0 / (b - a)) * coeff
```

Listing 2: Digital option COS payoff coefficients

8 Criteria Checklist

Table 4: W4 deliverable status.

#	**Requirement**	**Status**
1	COS method with Merton characteristic function	
2a	COS vs exact Merton, error $< 10^{-8}$	**at** $N = 128$
2b	COS vs Monte Carlo (within $\pm 2\sigma$ bands)	
3	Kou double-exponential CF + Merton vs Kou smile	
4	Digital option pricing via COS (all 4 types)	

References

- Fang, F. & Osterlee, C.W. (2008). *A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions.* SIAM Journal on Scientific Computing, 31(2), 826–848.
- Merton, R.C. (1976). *Option pricing when underlying stock returns are discontinuous.* Journal of Financial Economics, 3(1–2), 125–144.
- Kou, S.G. (2002). *A Jump-Diffusion Model for Option Pricing.* Management Science, 48(8), 1086–1101.
- Cont, R. & Tankov, P. (2004). *Financial Modelling with Jump Processes.* Chapman & Hall / CRC.